Curtin University

SPAT5008 Applied GIS

Assignment 2

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Learning Outcomes

After completing this Assignment, you should be able to:

- Select and acquire imagery appropriate for the task.
- Analyse imagery to extract meaning.
- Transform "raw" data into geospatial intelligence.
- Construct and execute research hypotheses.
- Work collaboratively.
- Present your research in multiple ways.

CHOOSE ONE OF THE FOLLOWING TOPICS

ASSIGNMENT 2: ANIMAL MOVEMENT

Main skills required are taught in Tutorial 6.

In Tutorial 6 we explore the home range and utilisation distributions of one green tree snake (*Morelia viridis*) in northern Queensland. In this assignment you will utilise all the green tree snake data to build up sufficient replicates so that you can make some generalisations. For example, what is the average home range, the smallest, the largest (and so on) using the MCP method and the UD-kernel method? Additionally, you can refine utilisation distributions using the dynamic Brownian bridge techniques. Are there any areas of common use? How many snakes might be found in common use areas? How big is a common use area? Where are they?

It will require a literature review on animal tracking data with specifics about snakes and perhaps other reptiles. Methods will need to explain the techniques used for home range assessment and the dynamic Brownian bridge. Consider how best you can present this information to your readers through tables and maps.

Your work must be written up as a paper (i.e. concisely) and should be between 4000-6,000 words (no bigger please!). It should include a thorough literature review, a study area section, methods, results, discussion, conclusion and of course references. Figures must be of a high quality. You may include R code in an appendix. Data can be found on Blackboard.

Highly suggested reading:

Hart, K., Cherkiss, M., Smith, B., Mazzotti, F., Fujisaki, I., Snow, R., & Dorcas, M. (2015). Home range, habitat use, and movement patterns of non-native Burmese pythons in Everglades National Park, Florida, USA. *Animal Biotelemetry*, *3*(1). doi: 10.1186/s40317-015-0022-2. Available at: https://link.springer.com/article/10.1186/s40317-015-0022-2

NB: Suggesting reading does not mean you should copy it or rely on it for anything other than a sensible place to begin to break down the problem.